

Alison Ricardo

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EDUCATION

Open University

BSc (Hons) Open Degree — Computer Science & Mathematics Pathway

Final 2026 modules: Software Engineering · Computational Applied Mathematics

Feb 2024 – Present

Remote

University of Westminster

BSc (Hons) Computer Science — Year 1 completed before transferring to Open University

Sep 2022 – Apr 2023

London, UK

Relevant Coursework: Computer Systems Fundamentals, Software Development II

PROJECTS

Quantum Computing Fundamentals

[Github Repository](#)

Python · Qiskit · Qiskit Aer · Matplotlib · Jupyter [Completed]

- Engineered and documented a Jupyter notebook implementing **4 core** quantum algorithms; superposition, entanglement, Bell states, and Deutsch–Jozsa — derived from mathematical first principles
- Designed and statistically validated a quantum random number generator across **600 measurement** shots, demonstrating rigorous, evidence-based verification methodology directly applicable to quantitative research and testing environments
- Independently researched the mathematical overlap between quantum computing and real-time rendering, reflecting a self-driven, cross-disciplinary approach to applying theory

Limit Order Book Simulator

C++ [In Progress]

- Architecting a limit order book matching engine in C++ implementing price-time priority order matching, including partial fill handling across multiple resting orders
- Designing and implementing core order book data structures supporting order insertion, cancellation, and best bid/ask retrieval
- Building a basic market-making strategy on top of the engine, incorporating inventory-risk-based quote skewing to manage one-sided position risk

EXPERIENCE

The Tea and Biscuit Club

Jan 2025 – Mar 2025

Shop Manager

Arundel, UK

- Managed full day-to-day operations of a tea shop, stock management, cash handling, and daily reconciliation
- Redesigned brand website end-to-end and owned all social media content creation, driving a **30% increase** in website traffic and online order volume through front-end development, performance optimization, and content strategy
- Designed and produced product packaging from brief to print-ready artwork using Adobe Creative Suite & Procreate
- Technologies:** Wix, Adobe Photoshop, Illustrator, Capct, Canva, Procreate

Staffwear

Feb 2024 – Present

Web Developer

Remote

- Building and maintaining the company's website end-to-end development, implementing responsive, mobile-first interfaces across **15+ pages** and managing all content updates and performance improvements
- Leading all visual brand output alongside web work — garment designs, print-ready artwork, advertising materials, and consistent brand identity across digital and physical formats
- Adapting creative and web direction based on performance data and audience feedback
- Technologies:** HTML, CSS, JavaScript, Adobe Photoshop, Illustrator, InDesign, Canva, Procreate

Sainsbury's

Jan 2024 – Jan 2025

Online Assistant

Chichester, UK

- Maintained high consistent targets across every shift - averaging scores of **220 items/hour**
- Worked across multiple store areas as required, demonstrating flexibility and reliability as a team member

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS, Java

Data Structures & Algorithms: Arrays, Trees, Graphs, Hash Maps, Dynamic Programming, Complexity Analysis

Quantitative / Scientific Computing: Qiskit, Jupyter, Matplotlib, NumPy, pandas, statistical verification methods

Mathematics: Linear Algebra, Probability & Statistics, Bayes' Theorem, Combinatorics

Tools: Git / GitHub, VS Code, Jupyter, Streamlit, Figma

Languages (spoken): English (native), Mandarin Chinese (learning)